Initial Proposal Form – 2025/26

Programme of Study:	Computer Science Bsc
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Proposed Title:

SpikeyProfile- Neurodivergence nfc & ai symptom and strength tracker and education tool

Proposed Aim:

I aim to develop a mobile application to help meet unique needs of people who are neurodivergent, especially with ADHD and Autism but focusing on more broad issues dealt with by neurodivergent individuals. The app would help to track symptoms such as mood, energy levels etc, help the user understand the potential benefits as well as the possible drawbacks of their neurotype and that these are contingent on their personal experience and possibly how to manage the inconvenient aspects of their conditions with non-medical solutions e.g suggesting a weighted blanket for insomnia or highlighting correlations between symptoms and other data the user provides such as facts about their environment or diet or activities with AI or providing a medication tracker that doesn't rely on memory as much for example optionally using nfc tags that can be attached to convenient locations(the wall above the medicine cabinet) and activated with a tap of a phone or nfc enabled accessory instead of having to manually log taking meds or completing stretches or other health tasks that users may struggle to track or have taxing workarounds for, and connect with a supportive community with signposts to forums for specific issues, symptoms or other disabled communities.

Rationale

I am convinced that this project needs to be undertaken because there is a gap in the availability of tailored support tools for neurodivergent people. Often, the one-size-fits-all approach in UI/UX design and functionality marginalizes neurodivergent people. For example, standard tools for tracking and managing medication typically require normal memory function, and those with severe memory issues usually rely on staff for assistance, however individuals with ADHD, who may struggle with memory but are otherwise independent, often find these tools inadequate. Most reminder apps offer a single secondary reminder, which is insufficient for users dealing with time-blindness and executive dysfunction. These users may need significantly more reminders and often resort to time-consuming workarounds, like manually setting multiple alarms.The app would also incorporate features like a clipboard tray that holds multiple items, user interface customization such as dark mode for users with light sensitivity ot filters for high contrast, colour-blindness or aesthetic pleasure – as the topic content health and wellbeing can be somewhat off-putting for those who need support in the area - and other enhancements to increase accessibility for users both within the app and in their daily lives. By addressing these gaps, the project aims to provide a comprehensive, customizable tool that empowers users to better manage their conditions and improve their quality of life.

Additionally, information is crucial for tracking and managing symptoms and helping users understand their unique needs. For instance, clarifying that Autism is a spectrum not a gradient (a common misconception), enables those with autism to gain a better insight and understanding of their unique experience allowing users to tailor their tracking to include specific sensory issues; as not all autistic individuals display all autistic symptoms and the cluster is usually unique, rather than a measure of intensity of a rigid set of symptoms. This can also include a record of reasonable accommodations, including suggestions backed with research and anecdotes to manage these symptoms effectively. The underlying idea behind this implementation is that high levels of personal customization would solve many problems with systems intended for use by neurodivergent individuals for tracking health information and understanding their unique neurotype over the long term with the use of the latest technologies. This project aims to be a novel software that follows the neurodivergent paradigm from the requirements collections stage throughout, which acknowledges that neurodivergence has its strengths as well as its weaknesses and takes a more holistic, person centred approach to technological solutions for users of all neurotypes. A spiky profile refers to the uneven distribution of cognitive abilities and skills where individuals demonstrate significant peaks and valleys in their capabilities, rather than consistent performance across different areas. Unlike a standard or "flat" neurotypical profile where abilities tend to cluster around a similar level with minor variations, people with spiky profiles exhibit pronounced differences between their strengths and weaknesses—for example, exceptional mathematical abilities paired with significant difficulties in social communication, or outstanding creative writing skills alongside struggles with basic organizational tasks. This cognitive pattern is predominantly associated with neurodivergent conditions including autism, ADHD, dyslexia, and dyspraxia, where the variation between peak abilities and challenging areas is much more dramatic than in neurotypical individuals. Research indicates that approximately 15% of the UK population is neurodivergent, with spiky profiles being a defining characteristic of these neuro-minorities. While all humans have some variation in their skills and abilities, the spiky profile represents a statistically significant disparity where abilities cross two or more standard deviations within normal distribution, making it a relatively common phenomenon among neurodivergent individuals but less frequent in the general population. This ability profile creates significant challenges in healthcare for neurodivergent people as they are not accounted for by standardised medical processes, creating communication problems leading to their needs being misinterpreted or overlooked in a situation commonly referred to as the “triple empathy problem”. Standard diagnostic tools, high-pressure clinical environments, and the tendency for neurodivergent individuals to mask their behaviors contribute to misdiagnosis, inadequate care, and systemic bias. Healthcare providers also frequently hold unconscious biases, leading to dismissive attitudes and inappropriate treatment. To address these problems, the healthcare system must adopt highly customizable tools and approaches—such as adaptive assessment methods, personalized care pathways, and modified environments—that honor individual cognitive differences. This shift toward precise person-centered care would elevate healthcare into a system that supports neurodiversity, improving both outcomes and patient experience. I intend for this app to be one such tool, a general implementation of the neurodivergent paradigm.